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Department of Informatics

Undergraduate Thesis

Developing a Mobile-Assisted Language Learning Application Utilizing Short-Form Video Reels and Hypercasual Games

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Abstract

Language Learning has become ...
Key words: [KEYWORD-1], [KEYWORD-2], [KEYWORD-3]
This thesis proposes a Personalized Language Learning (PLL) application that...

Περίληψη

Η παρούσα διπλωματική εργασία ...
Λέξεις κλειδιά: [ΛΕΞΕΙΣ-ΚΛΕΙΔΙΑ], [ΚΕΨΩΟΡΔ-1], [ΚΕΨΩΟΡΔ-2], [ΚΕΨΩΟΡΔ-3]

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I dedicate this work to my family.

Mohammad Matin Marzie
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Chapter A

Introduction

A.1 Motivation

The scale of foreign language acquisition is an unprecedented phenomenon in contemporary applied linguistics. In our globalised society, over 1.5 billion individuals are actively studying a foreign language [1], seeking to facilitate cross-cultural communication, academic mobility, and access to global scientific literature. This widespread interest in language learning in conjunction with the advent of technological advancement (e.g., smartphones, AI) has shifted the paradigm of second language acquisition (SLA) away from traditional classroom-bound instruction toward self-directed, digital learning environments. Currently, independent, mobile-assisted platforms represent the most widespread medium for independent language study, enabling millions of learners to engage with language materials on demand [2].

Despite the ubiquity of mobile-first access and digital resources, contemporary Mobile-Assisted Language Learning (MALL) faces critical pedagogical efficacy bottlenecks. While self-learning mobile applications have become the standard interface for independent L2 (second language) acquisition [2], their long-term educational utility is severely compromised by learner attrition. Industrial Analysis of self-directed language learning applications reveals a severe pedagogical retention deficit, with first-week user attrition rates exceeding 60% across many major platforms [2].

Furthermore, modern lifestyle demands and cognitive workloads compress the time available for structured, long-form study [3]. Traditional, exhaustive learning structures fail to integrate into the daily routines of adult learners, as evidenced by empirical findings indicating that 53% of active adult learners and half of educational directors identify excessive daily workloads as the primary barrier to ongoing instruction [3].To circumvent these temporal constraints, learner behaviors have increasingly shifted toward autonomous, bit-sized micro-learning formats.

While modern mobile applications have successfully popularised the micro-learning format, current digital implementations remain heavily constrained by structural and pedagogical bottlenecks. As analyzed in depth within the literature review (Section B), existing MALL platforms frequently rely on uniform, static curricula that fail to adapt to individual learner demographics, lack authentic multimedia integration, and reduce gamification to repetitive, decontextualized translation drills—a phenomenon termed "shallow gamification." Consequently, these platforms fail to support advanced proficiency development or sustain long-term cognitive engagement. This systemic deficit shifts the scientific inquiry: *how can mobile language learning frameworks transcend these repetitive, drill-based limitations to deliver adaptive, culturally authentic, and cognitively immersive media that aligns with the organic attention habits of modern learners?* Addressing this question requires investigating informal digital environments where contemporary learners willingly and habitually direct their attention.

A.2 Background

To address these completion and engagement deficits, modern Mobile-Assisted Language Learning (MALL) must align with the media consumption realities of contemporary learners. This thesis explores the integration of two prominent digital phenomena: short-form video and hypercasual game mechanics.

A.2.1 Short-form Video

Short-form video is a media format native to contemporary social media platforms, characterized by its brevity—typically ranging from 15 seconds to 3 minutes—and optimized for rapid, algorithmically driven consumption.

Short-form video has become the dominant interface of digital attention across social networks. Statistically, YouTube Shorts has achieved over 200 billion daily views [4], While Meta Platfrom announced: Video is a particular bright spot, with video time spent on Instagram up more than 30% since last year and 5% more time spent on Facebook [5]. TikTok's global active advertising audience has reached 1.59 billion adults, representing 27.5% of the global population aged 18 and older [6].

This media format captures a disproportionate share of mobile attention. Globally, TikTok Android users average 1 hour and 37 minutes per day in the application, Youtube 1H and 25M, Instagram 1H and 13M and Facebook 1H and 7M [7].

This behavioral pattern is heavily concentrated among younger cohorts: 75% of US teens visit YouTube daily, and 61% engage with TikTok daily [8]. Among US adults, daily TikTok usage exhibits a sharp ten-to-one generational gap, captured by 50% daily usage among 18-to-29-year-olds compared to just 5% among those aged 65 and older [9].

Pedagogically, the cognitive appeal of short-form video lies in its brevity and high visual density. While platforms keep raising video length caps (with TikTok allowing up to 10 minutes and Reels up to 3 minutes) [10], consumer behavior remains heavily anchored around rapid completion. Wistia's benchmark analysis of 13 million videos reveals that videos under 60 seconds hold a 52% average engagement rate, meaning viewers watch more than half the runtime before scrolling [11]. In contrast, longer videos see completion rates decay exponentially [11, 12]. Short-form videos therefore represent a highly engaging, authentic delivery mechanism for Second Language Acquisition (SLA).

A.2.2 Hypercasual and Hybrid-Casual Games

Hypercasual video games are defined as highly accessible, mobile-native software products characterized by minimalist user interfaces, instantaneous "tap-to-play" mechanics, and extremely short core gameplay loops [13]. Unlike casual games, which require some level of long-term planning, tutorial acquisition, or system familiarity, hyper-casual games demand no pre-existing cognitive schema or learning curve from the user, relying on immediately intuitive controls (core game mechanics) such as direction and drawing, matching, puzzle, risign and falling, swerving, swiping, tapping/timing and word and board [13, 14]. In recent years, this genre has evolved into "hybrid-casual" gaming, which retains the low-barrier entry and core mechanics of hypercasual games while strategically incorporating mid-core elements, such as progressive meta-features, item collection, and light narrative layer systems [14].

The exceptional appeal and potential addictiveness of these games are central to their design and rapid market expansion [14]. Reflecting this widespread appeal, the hyper-casual and hybrid-casual genres have experienced exponential audience expansion globally, emerging as a dominant leisure activity on both major mobile platforms [15].The addictiveness of these games are historically highlighted. Flappy Bird, which was voluntarily withdrawn from download by its creator due to concerns over its addictive and habit-forming impact on players [14].

Furthermore, because they are playable primarily on ubiquitous mobile devices and utilize intuitive user interfaces, hypercasual games reach a broad demographic spectrum. Notably, they appeal strongly to populations that historically do not identify as gamers or have been marginalized by the "male gamer stereotype" [16]. These include female players—who comprise 63% of the mobile gaming market [17]—as well as older adults (aged 65+) seeking cognitive maintenance, mental health preservation, and familial bonding [18].

Despite this high volume of daily engagement, players primarily consume these games as simple digital leisure pursuits to pass time, alleviate boredom, or fill transitional moments during daily routines and commutes [13, 14].The highly engaging, low-friction mechanics of hypercasual and hybrid-casual games represent a powerful, untapped attention-delivery system. While their mass appeal stems precisely also from this cognitively empty activities, ... TODO: Add here a quesiton, read yang2020hyper maybe to get help.

A.3 Problem Statement and Research Questions

People play a lot Hypercasual games, can they be leveraged to learn a language? How can we turn reels and games into language learning and make it effective? We have to track each word.(personalize) We have to Engineer a Reel recommendation system(algorithm) with personalized words. Personalize each Hypercasual game level.

The central research problem is the absence of a language learning system that simultaneously leverages (i) the authentic contextual richness of user-generated short-form video and (ii) gamified reinforcement adapted to the individual learner's vocabulary state.

Research questions:

RQ1. How can a mobile application integrate authentic short-form video content with a word-level vocabulary tracking model to support second language acquisition?

RQ2. How should a content recommendation engine be designed to select reels that are both comprehensible and lexically challenging for a given learner, based on forgetting curves?

RQ3. To what extent can hypercasual game mechanics, when parameterised by the learner's vocabulary state, serve as effective spaced-repetition exercises within a language learning application?

The question is: Can't i learn language by watching Reels from existing social media platforms? The answer is in next section **??**.

This thesis proposes **Glosy**, a cross-platform mobile application that addresses the identified gaps by embedding vocabulary acquisition within short-form authentic video content and reinforcing it through personalised hypercasual games. The system tracks each learner's vocabulary state at the word level, uses that state to select contextually appropriate reels, and adapts game difficulty to the same vocabulary model.

A.4 Significance

Helps learners use their daily habits for productive learning. The application is mobile-first. Supports Offline game play and practice. 3/4 use offline mode [19].

A.5 Thesis Structure

The remainder of this thesis is organised as follows. Chapter B reviews mobile-assisted language learning platforms, identifies their limitations, and establishes the pedagogical framework. Chapter C presents the system architecture, technology stack, data model, and recommendation engine design. Chapter **??** documents the implementation of each component with reference to the source code and database schema. Chapter E evaluates the system against its design objectives and discusses performance. Chapter **??** states the contributions, acknowledges limitations, and proposes future work.

Chapter B

Literature Review

B.1 The Landscape of Mobile-Assisted Language Learning (MALL)

B.1.1 Market Evolution and Platform Proliferation

Digital language learning platforms (LLPs) have become dominant in foreign language education, with over 70 major commercial applications serving millions of active learners through Mobile-Assisted Language Learning (MALL) [20]. The global market for digital language platforms was valued at USD 5.8 billion in 2025 and is projected to reach USD 21.1 billion by 2034, expanding at a Compound Annual Growth Rate (CAGR) of 15.43%, with Duolingo commanding over 50% of global downloads [19].

This rapid expansion has been further accelerated by Generative Artificial Intelligence and Large Language Models (LLMs), which have streamlined content authoring and curriculum scaling. While authoring the first 100 language courses on Duolingo required over 12 years of manual pedagogical design, generative models facilitated the deployment of 148 additional courses in early 2025 alone [21].

B.1.2 Critical Review and Systematic Limitations of Current Platforms

Karasimos [20] conducted a systematic cross-platform evaluation of five representative LLPs—Duolingo, Memrise, Busuu, Rosetta Stone, and LingQ—evaluating their pedagogical efficacy and interactive design. While these applications exhibit strength in habit formation and foundational syntax acquisition, a critical analysis reveals structural shortcomings across contemporary commercial architectures:

1. **Uniform Content Delivery Across Learner Demographics:** Existing platforms deliver static, uniform curricula that do not account for variations in age cohorts (e.g., Gen Z, adults, seniors), communicative objectives, or cultural nuances across non-English language pairs. Machine learning implementations in mainstream platforms focus predominantly on tuning exercise difficulty rather than personalizing semantic context [19].

2. **Insufficient Multimodal Integration:** Primary instructional routines rely predominantly on isolated text prompts and synthetic audio translation drills. This structure underutilizes visual and paralinguistic memory channels that reinforce natural conversational patterns and prosody.

3. **Absence of Authentic, Real-World Linguistic Context:** Mainstream curricula frequently isolate vocabulary into synthetic sentences. While LingQ incorporates real-world texts, its interface poses steep onboarding barriers for novice learners [20]. Authentic input is vital for communicative competence [22]; however, uncurated native materials risk cognitive overload if cultural and lexical scaffolding are missing [23].

4. **Superficial Gamification vs. Deep Mechanics:** Gamification in LLPs is largely restricted to extrinsic feedback mechanisms (e.g., XP points, daily streaks, leaderboards) that do not directly stimulate deep linguistic recall.

5. **Limited Advanced Proficiency Support (B2+):** Current mobile platforms focus heavily on A1–B1 grammar foundations, offering sparse resources for upper-intermediate and advanced learners requiring idiomatic fluency, register variation, and natural speech rates.

6. **Absence of Sociocultural Grounding:** Mainstream platforms frequently isolate grammar rules from their social contexts, abstracting linguistic mechanics from the cultural dynamics highlighted by Sociocultural Theory.

B.1.3 Comparative Synthesis of Commercial LLPs vs. Glosy

Table B.1 outlines the architectural and pedagogical differences between established commercial LLPs and the proposed *Glosy* platform.

Table B.1: *Comparative feature matrix: Established LLPs vs. Glosy framework.*

Feature / Dimension	**Duolingo**	**Memrise**	**Busuu**	**LingQ**	**Glosy**
Authentic Multimodal Video	Low	Medium	Low	Medium	**High**
Deep Gameplay Mechanics	Low (Extrinsic)	Low	Low	None	**High**
Adaptive SRS Tracking	High	High	Medium	High	**High**
Sociocultural Scaffolding	Low	Low	Medium	Medium	**High**
B2+ Proficiency Support	Low	Low	Medium	High	**High**
Automated Transcript Parsing	Low	Low	None	Medium	**High**

B.2 Pedagogical and Cognitive Foundations

B.2.1 Sociocultural Theory and the Zone of Proximal Development

Vygotsky's Sociocultural Theory posits that cognitive development occurs through socially mediated communication and cultural artifacts, wherein language functions as the primary cognitive tool. Central to this paradigm is the **Zone of Proximal Development (ZPD)**: the gap between a learner's independent competence and their potential development when assisted by structured scaffolding. Instructional design remains optimal when educational stimuli avoid cognitive oversimplification without exceeding learner comprehension.

In the *Glosy* architecture, the ZPD is operationalized computationally through a lexical comprehensibility threshold. A short-form video unit is recommended when the user's historical vocabulary profile covers at least 98% of the target transcript tokens—a metric aligned with lexical threshold models for unassisted comprehension. The remaining $\sim$2% constitutes an "$i + 1$" linguistic challenge, contextualized within authentic communicative scenarios.

B.2.2 Cognitive Load Theory and Microlearning Frameworks

Cognitive Load Theory (CLT) states that working memory capacity is strictly bounded, requiring instructional systems to minimize extraneous cognitive load to facilitate schema construction in long-term memory.

To mitigate cognitive fatigue, modern educational technologies leverage mobile **microlearning** structures. Currently, over 72% of instructional organizations utilize bite-sized learning frameworks to accommodate divided attention spans [24, 25]. Structured microlearning modules—typically lasting between 3 to 10 minutes [26]—reduce cognitive overhead and harness the spacing effect, increasing knowledge retention by 25% to 60% relative to uninterrupted long-form instruction [27].

B.2.3 Memory Decay and Spaced Repetition Models

B.2.3.1 The Ebbinghaus Forgetting Curve

Ebbinghaus demonstrated that mnemonic decay follows an exponential decay trajectory in the absence of spaced reinforcement. Spaced Repetition Systems (SRS) counteract this decay by dynamically scheduling vocabulary review immediately before predicted memory consolidation failure.

In *Glosy*, this mechanism is operationalized within the `user_vocabulary` datastore through a 6-tier parametric mastery model. Every lexical entry tracks an explicit `mastery_level` ($1 \leq m \leq 6$) coupled with a `last_review` timestamp to dynamically schedule target review intervals.

B.2.3.2 Half-Life Regression and Retention Tracking

To predict memory state transitions accurately, parametric half-life regression (HLR) models estimate the probability p of successful recall as a function of the time elapsed Δt since the last exposure and the word's current memory half-life h:

$$p = 2^{-\frac{\Delta t}{h}} \tag{B.1}$$

By integrating HLR-based scoring within the video recommendation and game-generation pipelines, *Glosy* schedules lexical items to reinforce retention across both consumption and gameplay loops.

B.3 Modalities for Acquisition and Active Reinforcement

B.3.1 Short-Form Video in Second Language Acquisition (SLA)

Authentic audio-visual materials provide a dense multimodal stream combining spoken phonology, pitch inflection, kinesics, facial cues, and sociocultural context. However, passive video consumption on social networks (e.g., TikTok, Instagram Reels, YouTube Shorts) presents an instructional paradox: high user engagement often yields poor long-term retention due to the absence of active cognitive retrieval. Without structured retrieval opportunities, passive lexical exposure remains subject to standard Ebbinghaus decay.

Glosy mitigates this dynamic by treating each video stream as an input channel tied directly to a persistent learner lexicon. Vocabulary encountered within a video reel is tagged and subsequently routed to active evaluation loops within the application's gameplay framework.

B.3.2 Deep Gamification: Hypercasual Mechanics for Active Recall

Rather than relying on superficial badge mechanics, deep gamification tightly couples game completion criteria with target pedagogical objectives. Hypercasual and hybrid-casual mechanics—characterized by intuitive touch interactions, minimal onboarding latency, and rapid core loops—serve as ideal vehicles for active linguistic retrieval [14, 13].

Glosy integrates two primary hypercasual gameplay modalities:

- **Lexical Deduction (Wordle Variant):** Learners reconstruct target vocabulary through iterative orthographic validation, drawing target words directly from recently encountered video transcripts.
- **Combinatorial Anagram Grids (Word of Wonders Variant):** Learners construct valid target words from a fixed set of phonemes/letters, forcing active orthographic retrieval rather than passive visual recognition.

B.4 Algorithmic Frameworks for Personalized Recommendation

B.4.1 Content-Based Filtering via Lexical Profiling

The content-based recommendation subsystem analyzes subtitle transcripts generated from short-form video units. Each transcript is tokenized, normalized, and mapped against the user's documented knowledge state in the `user_vocabulary` table. The candidate reel scoring function prioritizes videos that satisfy the pedagogical constraint:

$$\text{Coverage}(U, R) = \frac{|V_R \cap V_U|}{|V_R|} \geq 0.98 \tag{B.2}$$

where V_R is the unique set of lexical tokens in reel R, and V_U represents the known vocabulary lexicon of user U.

B.4.2 Collaborative Filtering for Cohort-Based Difficulty Calibration

Collaborative filtering constructs user-item interaction matrices based on playback completion metrics, word lookup frequencies, and downstream game accuracy across peer learner cohorts. This identifies engagement and difficulty patterns that pure lexical indexing cannot detect.

B.4.3 Hybrid Pedagogical Recommendation Architecture

The unified recommendation engine integrates content-based lexical filtering, collaborative behavioral patterns, and HLR memory-decay timers into a multi-objective ranking function:

$$S(U, R) = w_1 \cdot \text{ZPD\_Fit}(U, R) + w_2 \cdot \text{SRS\_Urgency}(U, R) + w_3 \cdot \text{CollaborativeScore}(U, R) \tag{B.3}$$

This formulation ensures that recommendations balance conversational authenticity, immediate comprehensibility $(i + 1)$, and overdue vocabulary reinforcement.

Chapter C

System Architecture & Methodology

C.1 System Overview

Glosy is designed as a three-service architecture in which a mobile frontend, a primary REST API backend, and a dedicated reels microservice each handle a well-defined subset of the system's responsibilities. Figure **??** illustrates the high-level interaction between these components. All services share a single PostgreSQL database instance and are coordinated in development via Docker Compose.

- **Frontend:** A cross-platform mobile application built with React Native (Expo SDK 54), targeting Android and iOS from a single JavaScript/TypeScript codebase.
- **Backend API:** A Node.js/Express service (ESM modules) responsible for user authentication, vocabulary management, progress synchronisation, and the dictionary.
- **Reels Service:** A Python FastAPI microservice Recommendation System responsible for serving reels enriched with structured dialogue data—sentences, timestamped tokens, and translations.
- **Database:** PostgreSQL 16, whose schema encodes the vocabulary model, user progress, content metadata, and engagement signals.

C.2 Technology Stack

C.2.1 Frontend — React Native / Expo

React Native was selected to enable cross-platform deployment from a single codebase without sacrificing the native rendering performance required for smooth video playback. Expo SDK 54 provides managed native modules for video (`expo-video`), secure credential storage (`expo-secure-store`), and build tooling (`eas-cli`). The application uses Expo Router (file-system-based navigation) and React Native Reanimated for gesture-driven animations in the game interfaces.

C.2.2 Backend — Node.js / Express

The backend is implemented as an ES-module Node.js application using Express 4. This choice reflects the project's existing expertise in JavaScript and the maturity of the Express ecosystem for REST API development. Authentication is handled with JSON Web Tokens (JWT): a short-lived access token (15 minutes) and a long-lived refresh token (30 days) follow the standard rotation pattern. Google OAuth 2.0 is supported alongside email/password authentication via the `google-auth-library`. Input validation uses Joi schemas for every endpoint, and the API is documented with Swagger (OpenAPI 3.0) accessible at `/swagger`.

C.2.3 Reels Service — Python / FastAPI

The reels microservice is implemented in Python using FastAPI and SQLAlchemy (async). The separation from the main backend is motivated by the richer data-processing requirements of the reels pipeline: joining reels with their associated dialogues, sentences, per-token word data, and sentence translations in a single query is better expressed in SQLAlchemy's ORM than in raw SQL via the Node.js `pg` client.

C.2.4 Database — PostgreSQL 16

PostgreSQL was chosen for its support of complex relational queries, array operators, and procedural functions—all of which are used in the vocabulary bulk-initialisation and the planned recommendation engine. The full schema is maintained in `database/thesis_db.sql`.

C.3 Data Modelling

The database schema models the domain along three axes: linguistic content, user state, and engagement signals.

Linguistic content

User state is captured in two tables. `user_languages` records each language pair a user is studying and `user_vocabulary` is the core of the learner model.

Engagement signals are stored in `reel_interactions`, which records views, likes, saves, comments, shares, and reports per (user, reel) pair. These signals are reserved for the personalised recommendation pipeline described in Section C.4.

C.4 Recommendation Engine Design

The recommendation engine is the theoretical core of the system. Its goal is to select, from the pool of available reels, those that are simultaneously *comprehensible* (the learner knows enough vocabulary to follow the content) and *pedagogically productive* (the reel contains words that are due for review or represent the next developmental step). The design follows a three-stage pipeline.

C.4.1 Stage 1: Comprehensibility Filtering

For each candidate reel, the engine computes the proportion of the reel's unique word tokens that are present in the learner's vocabulary model at mastery level ≥ 1. A reel passes the filter only if this proportion meets or exceeds 98%, corresponding to Nation's [nation2001learning] threshold for unassisted reading comprehension. This criterion implements the ZPD operationally: the learner encounters real language while remaining within a comprehensible range.

C.4.2 Stage 2: Spaced-Repetition Scoring

Each reel is scored according to the review urgency of the vocabulary it contains. For each word w in the reel with mastery level m and days since last review d, the system computes an urgency score based on the Ebbinghaus forgetting function. The reel's aggregate score is the mean urgency across its target vocabulary. This stage ensures that reels containing words the learner is about to forget are ranked above reels containing words that were recently reviewed.

C.4.3 Stage 3: Content Personalisation

The final stage personalises the recommendation by incorporating engagement signals. Reels that have been liked are boosted. This stage allows the system to learn the learner's preferences over time, promoting content that is both pedagogically useful and personally engaging, thus considering demographic factors too.

C.5 Gamification Design

The gamification layer consists of two components: a progression system and two vocabulary-driven games.

The **progression system** (XP), (Coins) provide a secondary reward currency redeemable for in-app benefits.

C.6 Vocabulary-Driven Games

C.6.1 Wordle-style Game

C.6.2 Word of Wonders

Chapter D

Implementation Details

D.1 Database Schema

D.2 Backend API

The API has been documented using Swagger/OpenAPI. The documentation is available at `/api-docs` endpoint.

D.2.1 Authentication Module

The backend (`backend/`) implements a dual-token authentication scheme. On successful login or registration, the server issues a short-lived JWT access token (default: 15 minutes) and a long-lived refresh token (default: 30 days). Google OAuth 2.0 is also supported.

D.2.2 Registration and Vocabulary Initialisation

D.2.3 Vocabulary Synchronisation

D.3 Frontend Architecture

The frontend has 5 tabs. The application implements an offline-first strategy by caching API responses in asyncStorage 3/4 use offline .

D.4 Reels Service

D.4.1 Service Architecture

The reels microservice (`reels-service/`) is a FastAPI python application.

D.4.2 Creating Reels

D.4.3 Dialogue Enrichment

Each uploaded reel response embeds the full dialogue associated with the reel's video content.

D.4.4 Personalisation Implementation Status

D.5 Game Modules

D.5.1 Wordle-style Game

D.5.2 Word of Wonders

Word of Wonders is a grid-based word-construction game in which the player must identify hidden words from a set of provided letters. Each game level is associated with a set of target words. In Glosy, these target words are drawn from the player's active vocabulary model, prioritising words at lower mastery levels. The grid is generated by computing all valid arrangements of the target words' letters into a crossword-style grid. pseudocode for the grid generation algorithm

Chapter E

Evaluation & Results

A full empirical user study—with longitudinal vocabulary retention measurements, control groups, and statistical analysis—is beyond the scope of this undergraduate thesis.

Chapter F

Conclusions & Future work

F.1 Contributions

F.2 Limitations

The work presented in this thesis has several limitations that must be acknowledged honestly. **No empirical user study.** The pedagogical effectiveness of the system has not been evaluated in a controlled experiment. **Limited content library.**

F.3 Future Work

The following directions are identified as high-value:

- **Automated content annotation pipeline:** A pipeline that ingests a raw video URL, transcribes speech using a large language model, aligns the transcript to video timestamps, tokenises and POS-tags the output, maps tokens to the word database, and produces a fully annotated reel record—reducing the per-reel curation cost from hours to minutes.
- **Empirical evaluation:** A randomised controlled study measuring vocabulary gain, retention at one week and one month, and learner engagement metrics, comparing Glosy against a no-game control condition and against Duolingo.

- **Multi-language expansion:** The schema and service layer are language-agnostic (language is a foreign key throughout); adding a new language requires only content ingestion, not code changes.
- **Collaborative learning features:** Shared vocabulary challenges, leaderboards scoped to friend groups, and comment-driven vocabulary discussion on individual reels, supporting the social dimension of Vygotsky's theory.
- **Vocabulary synchronisation performance:** Replacing the serial per-word UPDATE loop in `syncController.js` with a bulk `INSERT ... ON CONFLICT DO UPDATE` statement to support large offline sessions without latency degradation.

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Abbreviations

AI - Artificial intelligence

Glossary

Algorithm - A sequence of instructions for solving a problem